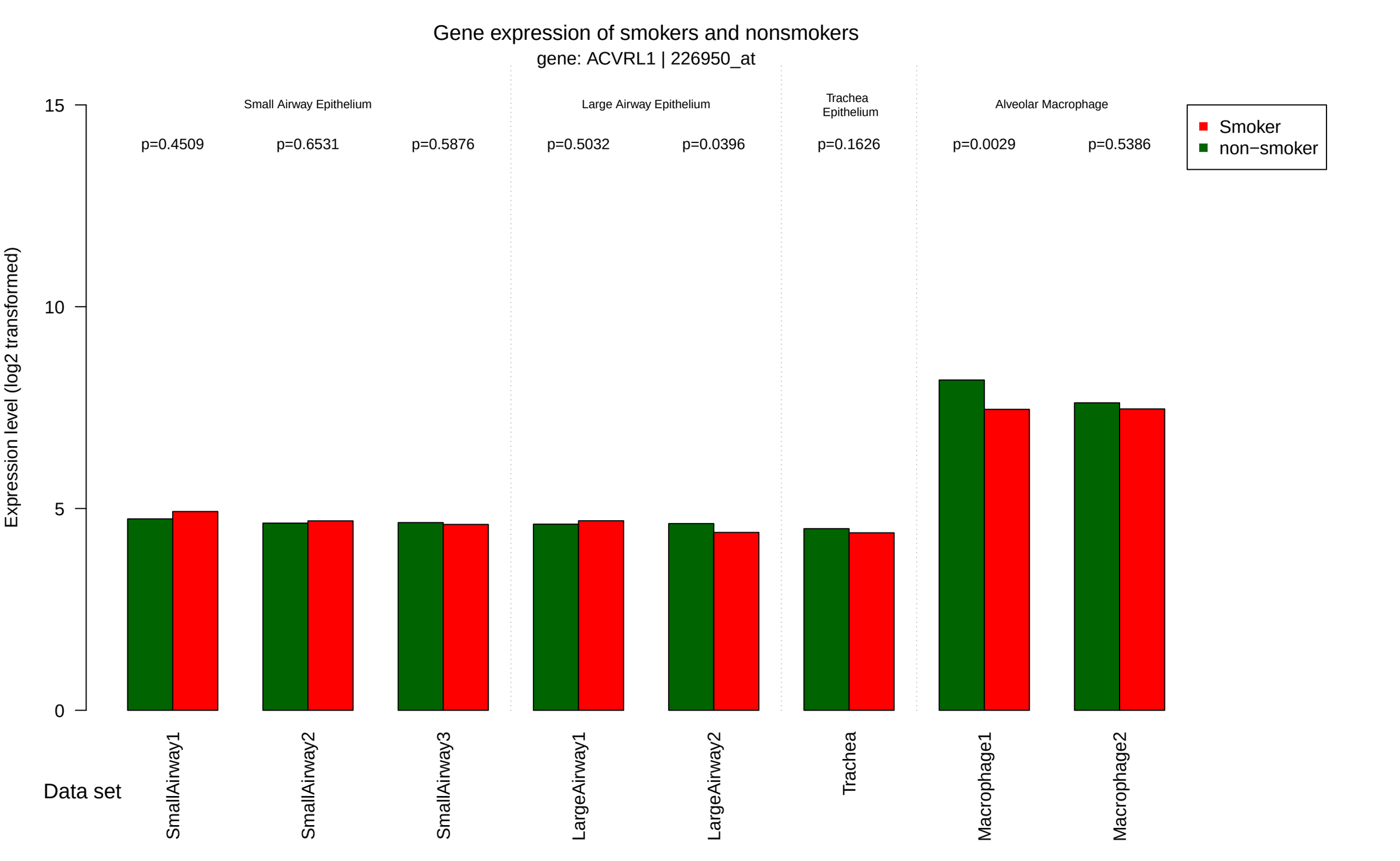
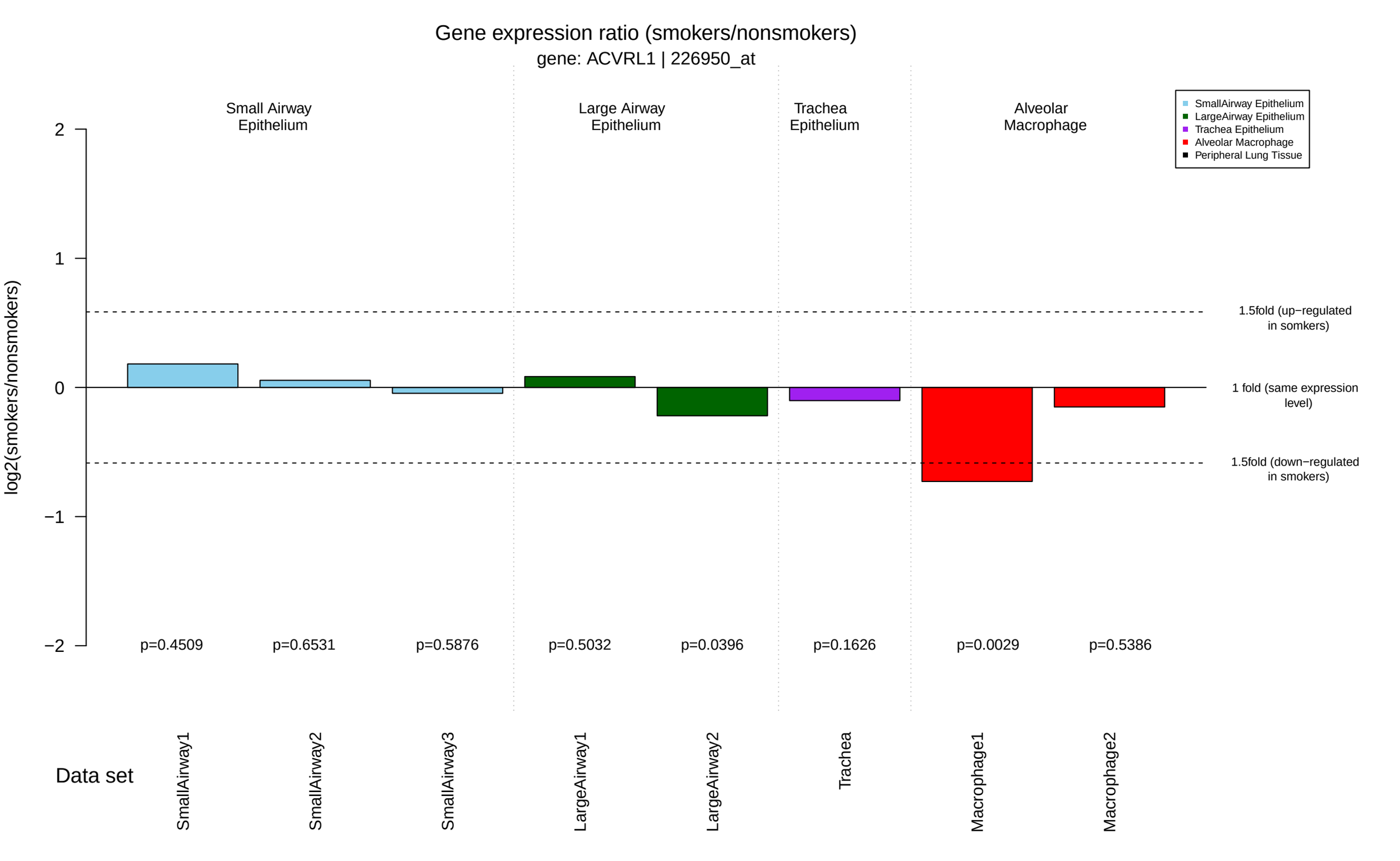
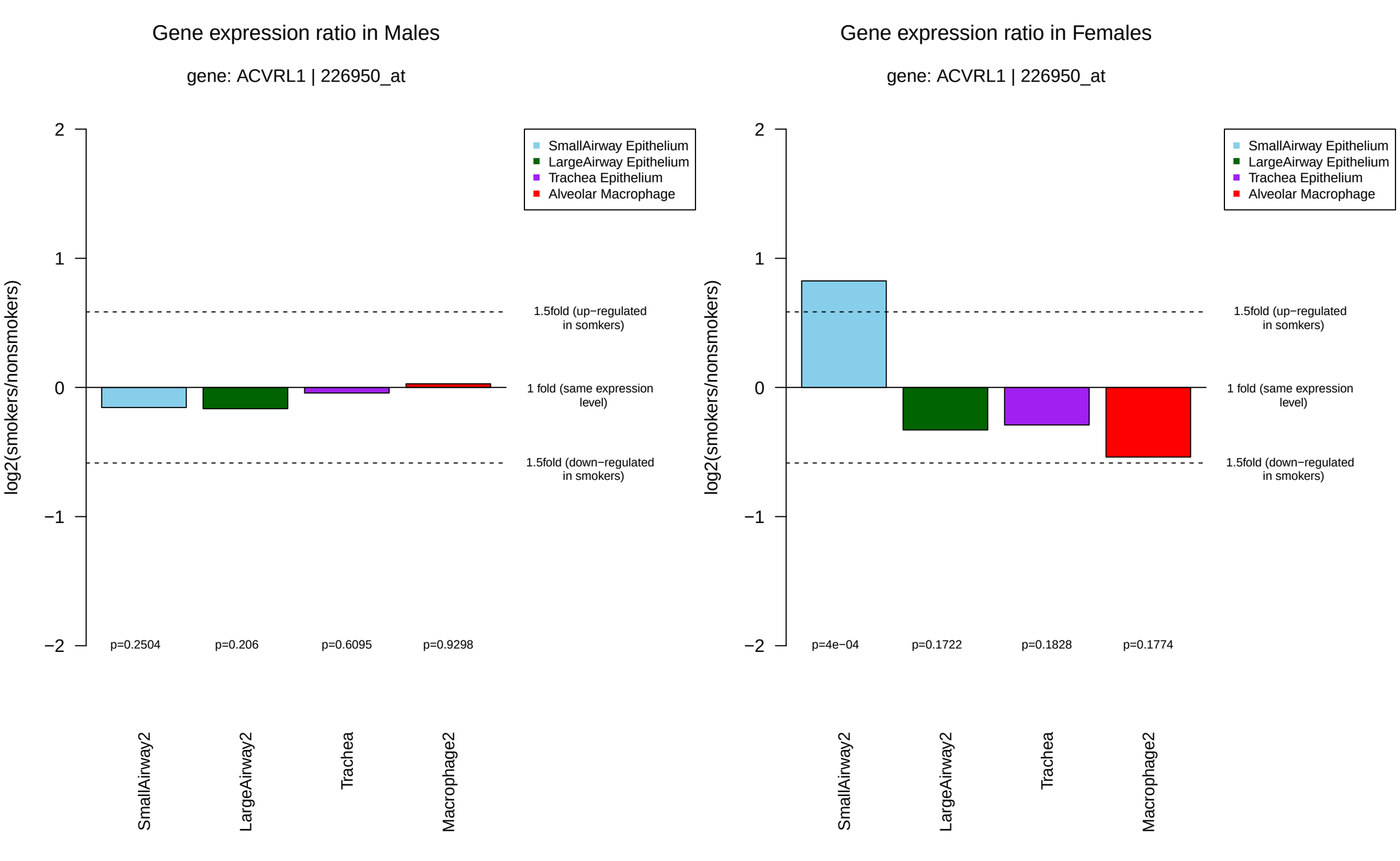
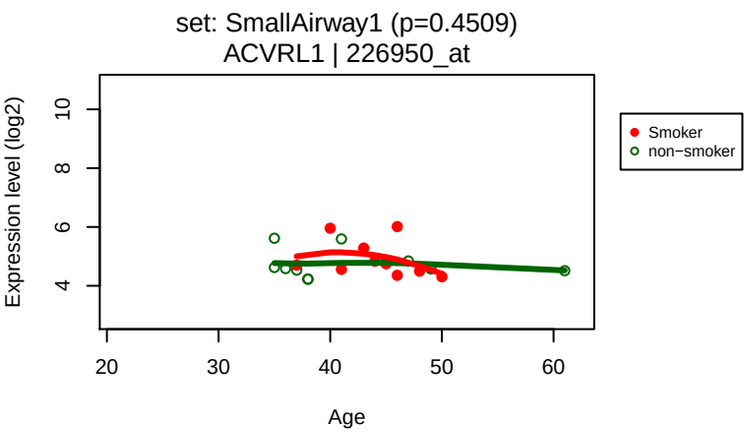
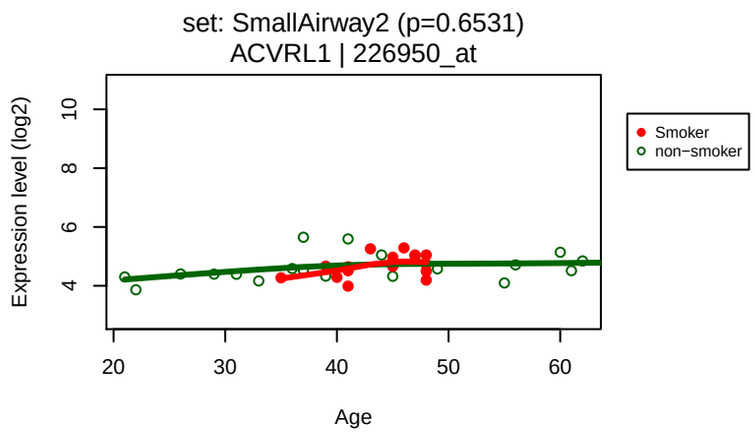
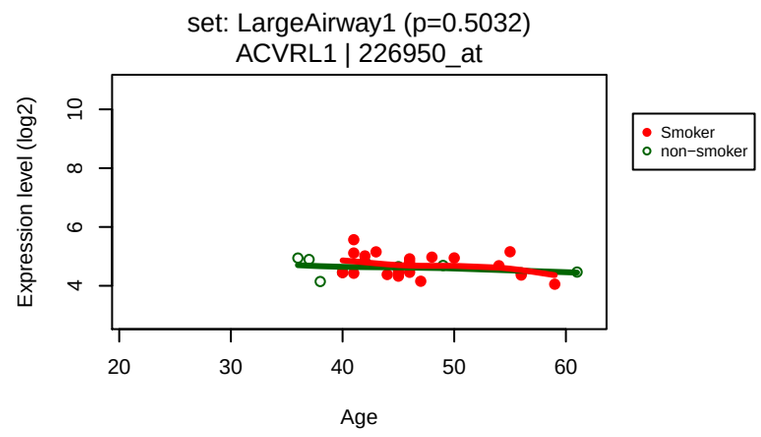

Gene expression of smokers and nonsmokers
gene: ACVRL1 | 226950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4509
p=0.6531
p=0.5876
p=0.5032
p=0.0396
p=0.1626
p=0.0029
p=0.5386
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

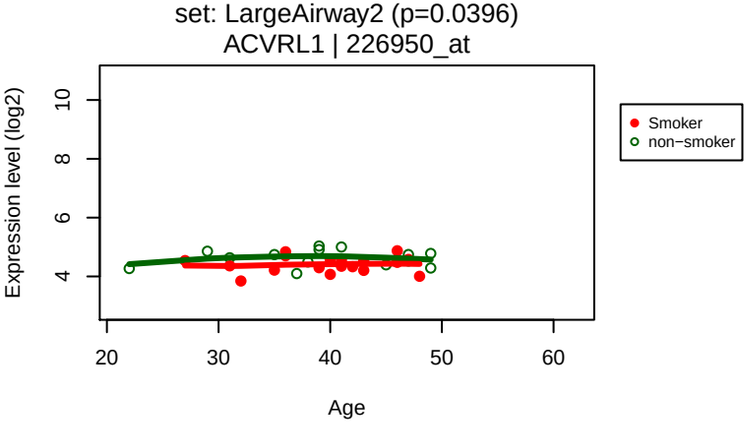

Gene expression ratio (smokers/nonsmokers)
gene: ACVRL1 | 226950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4509
p=0.6531
p=0.5876
p=0.5032
p=0.0396
p=0.1626
p=0.0029
p=0.5386
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

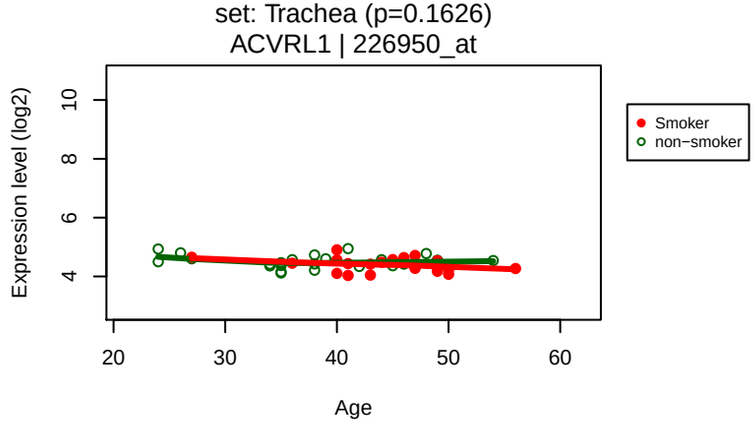

Gene expression ratio in Males
gene: ACVRL1 | 226950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2504
p=0.206
p=0.6095
p=0.9298
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACVRL1 | 226950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=4e−04
p=0.1722
p=0.1828
p=0.1774
SmallAirway2
LargeAirway2
Trachea
Macrophage2

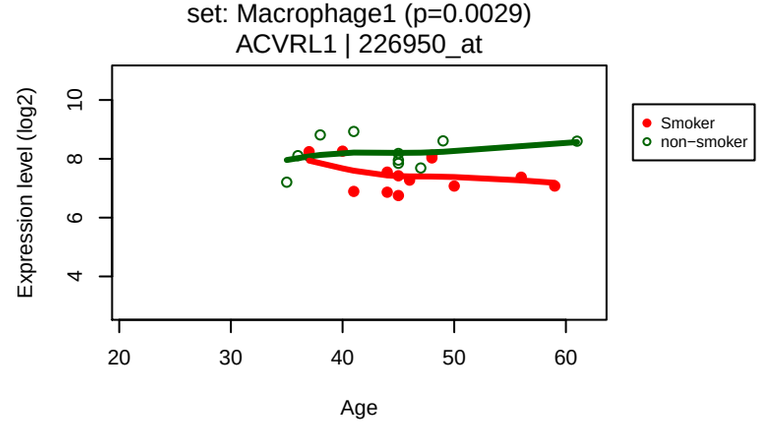

set: SmallAirway1 (p=0.4509)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker

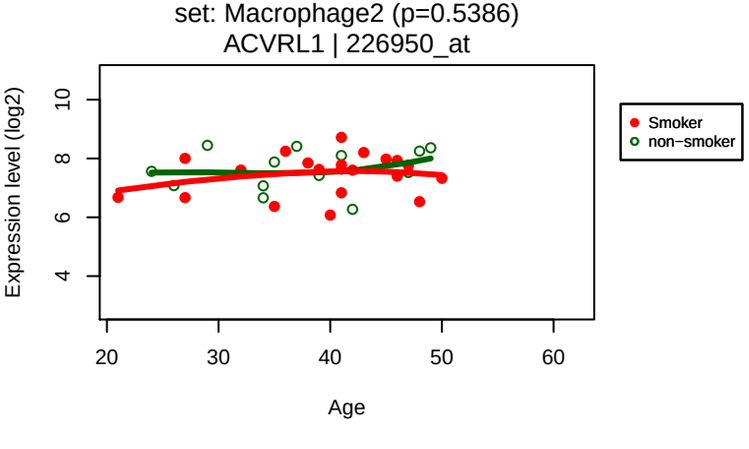

set: SmallAirway2 (p=0.6531)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker

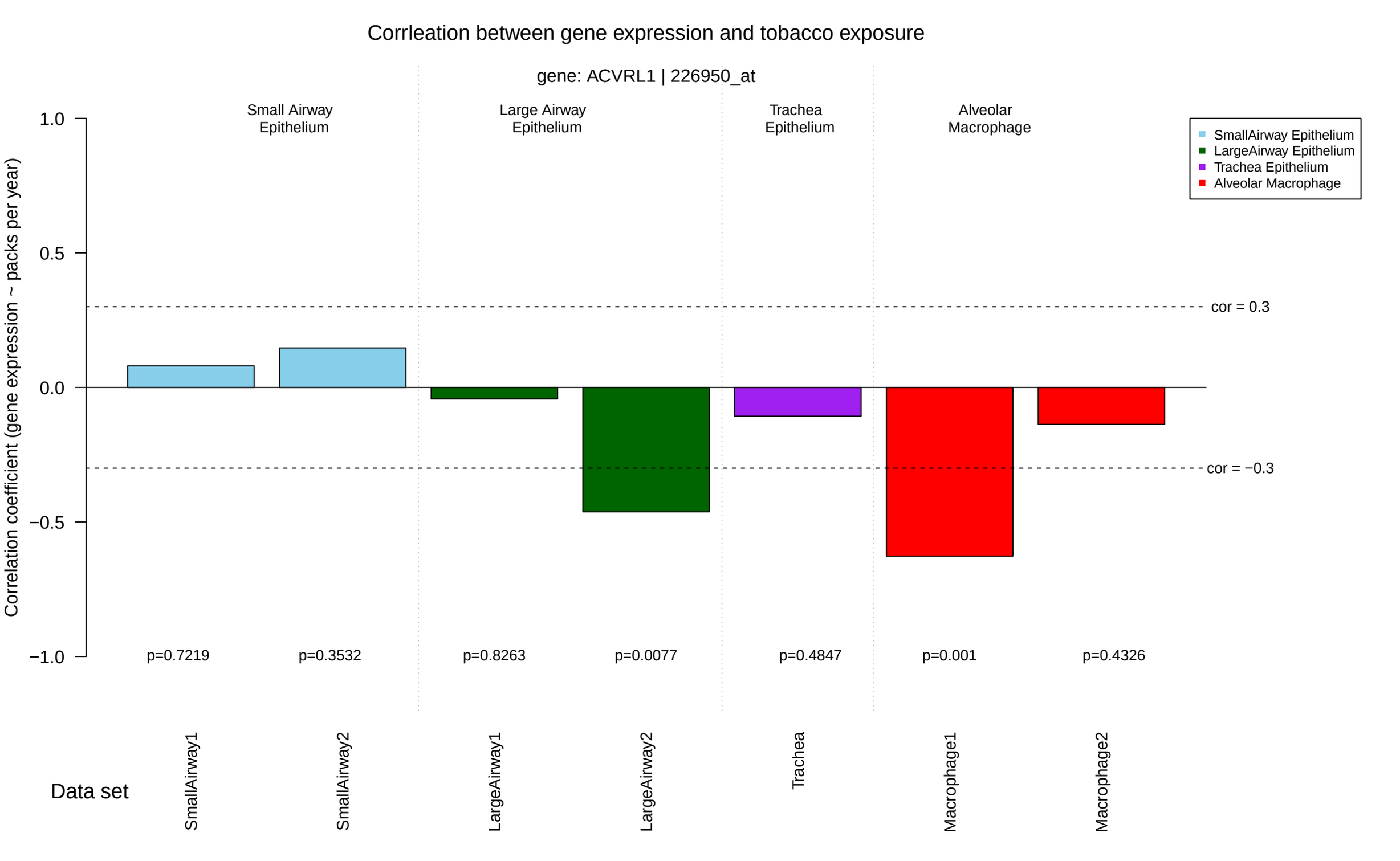

set: LargeAirway1 (p=0.5032)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0396)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1626)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0029)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5386)
ACVRL1 | 226950_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACVRL1 | 226950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7219
p=0.3532
p=0.8263
p=0.0077
p=0.4847
p=0.001
p=0.4326
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2